

Minor Project

Iris Flower Classification Overview:

The **Iris dataset** is a well-known dataset in machine learning used for classification. It consists of 150 observations of iris flowers from three different species: Setosa, Versicolor, and Virginica. The objective of this project is to predict the species of an iris flower based on its features.

Steps:

- **Dataset:** The Iris dataset includes features like petal length, petal width, sepal length, and sepal width.
- **Algorithms:** You can apply simple yet effective algorithms like **K-Nearest Neighbors (KNN)** or **Decision Trees** to classify the flowers based on the features. KNN classifies based on the majority label among the nearest neighbors, while decision trees build a model by splitting the dataset at each node based on feature values.
- **Evaluation:** After training the model, accuracy is measured to evaluate its performance. You can also use confusion matrices, precision, recall, and F1-score for more in-depth evaluation.

Application:

This project is perfect for beginners to get hands-on experience with classification algorithms and understand how models can learn patterns from data. The project introduces the key concept of supervised learning and classification.